

More common stuff

Here is a **concise checklist of common items** that *every other test* should include, based on your **sanity_test** structure and the TA's grading contract. Think of it as a **template of required steps** that should appear in all tests (with the test-specific stimulus in the middle).

Common Required Pattern (for every test)

1) Initialize and log

- Print a clear start line:

```
[INFO] <test_name>: starting
```

- This makes debugging and logs consistent.
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2) Keep BFM in lock-step with CTRL

Every test that sets CTRL fields **must also set**:

- `tb_top.bfm_mode`
- `tb_top.bfm_width`
- `tb_top.bfm_lsb_first`
- `tb_top.bfm_miso_word` (or at least `bfm_pattern`)

Rule:

If you change MODE/WIDTH/LSB_FIRST in CTRL, update the BFM signals in the same test.

3) Program registers in a valid sequence

Typical safe order:

1. `CTRL` (EN/MSTR/mode/width/lsb/loopback)
2. `CLK_DIV`
3. `DELAY`
4. `INT_EN`

5. `SS_CTRL` (assert SS before pushing TX)
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4) Predict before you push

Before every transfer (or group of transfers), call the ref model predictor:

- For 8-bit: `predict_single_byte( ... )`
- For 16/32: `predict_word( ... )`

This ensures your check has a correct expected value.

5) Send stimulus

- write TX_DATA
 - assert SS_n (SS_CTRL)
 - if multiple words: write TX_DATA multiple times while SS stays asserted
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6) Wait for completion (with timeout)

Always:

- poll `STATUS.BUSY`
 - break after some max loops
 - if timeout, log `[CHECKER_ERROR]` and increment error_count
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7) Read RX + check

- read RX_DATA
 - call `check_rx` or `check_rx_word`
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8) Deassert SS

Always clean up:

- `apb_write(SS_CTRL, 0)`
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9) Sample functional coverage

At least:

- `coverage.sample_config(mode, lsb_first, width)`
Other coverage bins should be sampled from test-specific fields (CLK_DIV, DELAY, IRQ, FIFO, etc.)
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10) Finish cleanly

- Print `[INFO] <test_name>: finished`
 - Do *not* print `[TEST_PASSED]` or `[TEST_FAILED]`
(tb_top does that)
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Extra common items for specific test classes

A) Register tests (`reg_access_test`)

- Must read reset values **before any writes**
- Must write/readback each RW register
- Must verify RO behavior (STATUS, RX_DATA)

B) Transfer tests (`mode_coverage_test` , `width_coverage_test` , `loopback_test`)

- Must update BFM config each time you change CTRL
- Must call predictor + check for every transfer

C) FIFO / interrupt tests

- Must include polling + explicit W1C actions
 - Must include *at least one* `[SCOREBOARD_ERROR]` on mismatch
 - Must avoid infinite loops
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Minimal “test skeleton” you should follow

```
[INFO] test_name: starting
set BFM config
program CTRL/CLK_DIV/DELAY/INT_EN
predict expected data
sample coverage
drive TX + assert SS
poll BUSY (with timeout check)
read RX and check
deassert SS
[INFO] test_name: finished
```
